

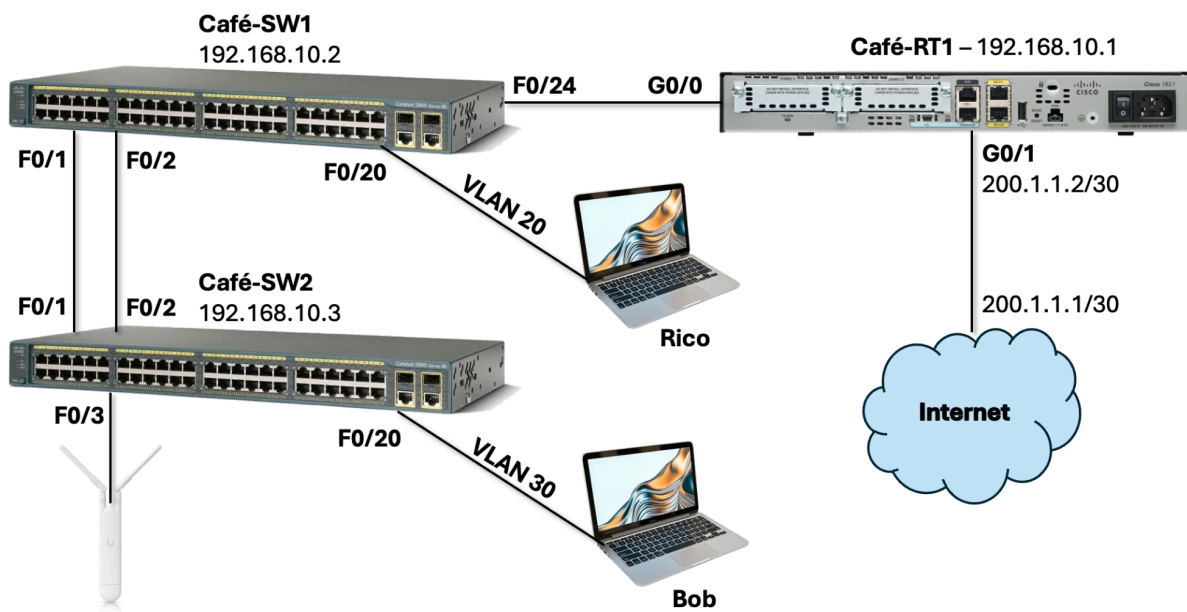


JULY 9, 2026

RULES OF THE GAME

- Configuration focus; light description of each concept is all that's permitted. Zach limits all conversation with the sacred squirrel.
- We will stop every 5 minutes* for any relevant questions. Micah will manage questions and pop them up on-screen for quick answers.
* = assuming we're not running out of time.
- The lab was created top-of-mind by a very non-AI Jeremy. If he forgot something and you call it out (seen by Micah), we'll implement it and give you a free Summer of CCNA T-Shirt. Also true if Jeremy makes a mistake and you help solve the problem (seen by Micah).

THE LAB



CONSOLE, FACTORY DEFAULT, & PHYSICAL CABLING

You just bought a glut of gray-market Cisco gear from EBay for your Cafe. Erase the configurations on all three devices and cable it to the previous network diagram.

IP ADDRESS MANAGEMENT (IPAM)

We will have three different VLANs at NetworkChuck Coffee shop:

- Management (VLAN 10)
- Administration (VLAN 20)
- Public WIFI (VLAN 30)

Break the 192.168.10.0/24 network into at least three subnets to accommodate this. Each subnet should have at least 50 hosts.

BASE CONFIGURATION

Time to get the base configuration going for all Cisco devices. Ensure each device has the following:

- Hostname
- Logon banner
- Exec secret password (priv. mode)
- Telnet password
- Console password
- Clear password encryption
- Port descriptions
- Management IP address (VLAN 10)
- Synchronous logging (console)
- Configuration saved to flash

VLAN & IP CONFIGURATION

Set up and name the VLANs on the two switches and set up a R0AS configuration to route between all three VLANs. Assign the router the first IP address from each subnet.

Configure trunk ports between switches, the WAP, and to the R0AS. Allow only the three VLANs specified for the Cafe. Set VLAN 10 as the native VLAN for the trunk ports. The WAP only gets VLAN 30 (and VLAN 10 native).

Create a “dead VLAN” (999) and assign all unused interfaces to the dead VLAN.

Set up DHCP scopes on the R0AS for each of the subnets that reserves the first 10 IP addresses of each range and hands out the remaining scope and necessary gateway. Use 1.1.1.1 as the DNS server and nccoffee.com as the domain.

Assign Rico’s laptop to VLAN 20 and Bob’s laptop to VLAN 30. Ensure they get IP addresses as expected (verify on the laptop AND on the router).

ROUTING & NAT CONFIGURATIONS

Set up the IP address 200.1.1.2/30 on the Gi0/1 interface of the router. Verify it can ping the ISP at 200.1.1.1.

Set up a static default route on the router that simply sends all unknown traffic out interface Gi0/1. Verify it appears in the routing table along with the local routes for the VLANs.

Configure NAT in such a way that ALL devices in the Cafe can share the public IP address. Verify all devices in the system can ping 200.1.1.1 (yes, even the switches). Verify NAT entries are shown on the router as devices ping.

SWITCHPORT CONFIGURATIONS

We've got some old school switches for the cafe (100Mbps should be fine, right?). Hard code speed and duplex on switch uplink ports (switch-to-switch connections).

View the CAM table to ensure MAC addresses are being learned on expected interfaces.

Make sure the switches see each other via CDP.

Clear the mac address table and verify addresses re-learn as expected.

STP & ETHERCHANNEL CONFIGURATION

Verify Spanning-tree configuration and identify what port(s) are blocked.

Configure all switches to use Rapid-PVST, then set up the Cafe-SW1 as the root bridge of the network. Verify your changes have taken effect.

Configure ALL ports except the trunk links to use Portfast and BPDUGuard.

Bundle the redundant ports into an Etherchannel using LACP negotiation. Change the load balancing of Etherchannel to consider source and destination MAC addresses in the algorithm.

CELEBRATE!!

We have configured the core concepts that power 99% of the networks of the world in one fantastic live stream.